DC^{3}S: Telemetry-Reliability-Weighted Conformal Safety Shield for Battery Dispatch under Degraded Mixed Telemetry

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Abstract

Battery storage dispatch under degraded telemetry exposes a critical safety gap: when sensor readings are delayed, dropped, or corrupted, uncertainty estimates become miscalibrated and optimization solvers issue unsafe commands. We introduce DC^{3}S (Detect–Calibrate–Constrain–Certify–Shield), a conformal safety shield that scores telemetry reliability at each control step, inflates prediction intervals via a Reliability-Adaptive Conformal Certification (RAC-Cert) law, projects candidate dispatch actions into the feasible safe set, and emits an auditable certificate. Evaluated on a CPSBench harness covering four fault dimensions (dropout, delay-jitter, spikes, out-of-order) across Germany (OPSD, 17,377 rows) and three US balancing authorities (EIA-930 MISO/PJM/ERCOT, ∼13,500 rows each), DC^{3}S achieves **zero true-SOC violations** with a 2.8% intervention rate while the deterministic baseline violates at 3.9% with P95 severity of 333 MWh. Cost-side, DC^{3}S-wrapped dispatch yields 7.11% cost savings, 0.30% carbon reduction, and 6.13% peak shaving on DE; stochastic value of the stochastic solution is positive in both regions (DE VSS = 2,708.61, US VSS = 297,092.71). A nine-row ablation with Wilcoxon signed-rank gates isolates RAC-Cert inflation as the critical safety component. All quantitative claims are locked to versioned run artifacts and validated by an automated claim-consistency checker.

Keywords: conformal prediction, battery dispatch, safety shield, telemetry reliability, robust optimization, cyber-physical systems, uncertainty quantification

1 Introduction

Grid-scale battery energy storage systems (BESS) are essential for integrating variable renewable generation, yet their dispatch depends on real-time telemetry—state-of-charge readings, power measurements, and grid-frequency signals—that is routinely degraded by sensor dropout, communication delay, spike-injection, and packet reordering [19, 16]. When telemetry quality drops, the prediction intervals that inform optimization widen unpredictably, and dispatch commands computed from stale or corrupted state can push the battery into unsafe SOC regions, risking equipment damage and grid instability [18].

Existing approaches address parts of this pipeline in isolation. Conformal prediction provides distribution-free coverage guarantees [1, 2], and adaptive variants track distributional shift [3, 4], but neither incorporates telemetry-quality signals into the calibration step. Robust and stochastic dispatch formulations protect against load/price uncertainty [7, 9], but assume the uncertainty sets themselves are well-calibrated—an assumption violated under degraded telemetry. Safety shields in control theory enforce state constraints [11, 12], but typically do not reason about measurement reliability or provide dispatch-level audit trails.

This paper bridges these gaps with DC^{3}S, a conformal safety shield that:

1. **Detects** telemetry degradation via a composite reliability score w_t;

2. **Calibrates** prediction intervals through RAC-Cert, inflating width proportional to degradation severity and drift risk;
3. **Constrains** dispatch by solving a robust optimization problem over the inflated intervals;
4. **Certifies** each dispatch action with a signed certificate recording the reliability state, interval widths, solver status, and constraint margins;
5. **Shields** the battery by projecting any infeasible candidate action to the nearest safe command.

Research questions.

RQ1 Does RAC-Cert inflation eliminate true-SOC violations under telemetry faults without excessive cost penalty?

RQ2 How does DC^{3}S compare to deterministic, fixed-interval robust, and CVaR dispatch baselines on the cost–safety Pareto frontier?

RQ3 Which shield components (reliability scoring, interval inflation, action repair) are necessary—and which are sufficient—for safety?

RQ4 Do safety and cost outcomes transfer across regions (DE $\leftrightarrow$ US) and seasonal regimes?

Contributions.

1. **RAC-Cert**: a reliability-adaptive conformal certification law that inflates intervals as a function of telemetry quality, drift risk, and sensitivity, with proved marginal coverage preservation (Theorem 1).
2. **CPSBench**: a fault-injection benchmark with truth-vs-observed SOC evaluation across four fault types and parametric severity sweeps.
3. **Empirical evidence**: zero-violation safety with quantified cost–safety tradeoffs on DE and three US balancing authorities, including ablation and transfer studies.
4. **Governance stack**: automated metric-locking and claim-validation tooling that prevents manuscript–artifact drift.

2 Related Work

Conformal prediction for time series. Conformal prediction [1] provides distribution-free prediction intervals with finite-sample coverage guarantees under exchangeability. Extensions to time-series settings include adaptive conformal inference (ACI) [3], which adjusts the miscoverage rate online, and conformalized quantile regression (CQR) [5], which wraps quantile regressors with conformal calibration. Regime-aware variants [6] stratify calibration by volatility bin. None of these methods condition calibration on *telemetry quality*, leaving a gap when input measurements are themselves degraded.

Robust and stochastic battery dispatch. Robust optimization for energy dispatch [7, 8] hedges against worst-case uncertainty within prescribed sets. Stochastic programming [9] optimizes expected cost over sampled scenarios. CVaR-based formulations [10] manage tail risk via Rockafellar–Uryasev linearization. These methods assume well-specified uncertainty sets; when the sets are miscalibrated due to sensor faults, the robustness guarantees degrade. DC^{3}S addresses this by coupling the uncertainty-set construction to telemetry reliability.

Safety shields and runtime monitors. Shield synthesis [11, 13] enforces safety by constraining the action space to pre-computed safe regions. Simplex architectures [14] switch to a safe baseline controller upon detecting anomalies. Runtime verification monitors [15] check trace properties post-hoc. DC^{3}S combines elements of all three: it monitors telemetry health, inflates the optimizer's constraint set, and projects actions to the safe set—while emitting a per-step certificate for auditability.

Telemetry reliability in CPS. Telemetry degradation in industrial control has been studied in the context of SCADA security [16] and anomaly detection [17], but integrating a telemetry-quality score directly into the dispatch optimization loop is, to our knowledge, novel.

Positioning. Table 1 summarizes the methodological positioning. DC^{3}S is the first method to combine conformal calibration, telemetry-reliability weighting, dispatch-level action repair, and per-step certification in a single framework.

Table 1: Methodological positioning of DC^{3}S relative to prior work.

Method	*Conformal*	*Telemetry-aware*	*Dispatch optim.*	*Action repair*	*Certificates*
CQR / ACI	✓				
Robust dispatch			✓		
Safety shields				✓	
Runtime monitors					✓
DC^{3}S (ours)	✓	✓	✓	✓	✓

3 Problem Formulation

3.1 Notation and Setting

Consider a discrete-time battery dispatch problem over horizon T with time step $\Delta t = 1\,\mathrm{h}$. At each step t, the controller observes telemetry $\mathbf{z}_t \in \mathbb{R}^d$ (load, wind, solar, price, SOC readings) and must select a dispatch action $a_t \in \mathcal{A}$ (charge/discharge power) subject to physical constraints.

Definition 1 (Telemetry degradation)**.** *A degradation operator $\mathcal{D} : \mathbb{R}^d \to \mathbb{R}^d$ maps clean telemetry $\mathbf{z}_t$ to observed telemetry $\tilde{\mathbf{z}}_t = \mathcal{D}(\mathbf{z}_t)$, where $\mathcal{D}$ may introduce dropout (zeroing), delay (time-shift), spikes (additive outliers), or reordering (permutation of recent samples).*

Definition 2 (Truth vs. observed SOC)**.** *Let SOC_t^{true} be the actual battery state governed by physics, and SOC_t^{obs} the value reported by (possibly degraded) sensors. The safety evaluation is on SOC_t^{true}; the controller only sees SOC_t^{obs}.*

3.2 Battery Dynamics

The true SOC evolves as:

$$\mathrm{SOC}_{t+1}^{\mathrm{true}} = \mathrm{SOC}_t^{\mathrm{true}} + \frac{\eta_c \cdot [a_t]^+ - [a_t]^- / \eta_d}{E_{\max}} \cdot \Delta t, \tag{1}$$

where $\eta_c, \eta_d \in (0, 1]$ are charge/discharge efficiencies, $E_{\max}$ is the rated energy capacity, and $[\cdot]^+, [\cdot]^-$ denote positive and negative parts.

The feasible action set enforces SOC bounds, power limits, and ramp constraints:

$$\mathcal{A}_t = \big\{a \in [-P_{\max}, P_{\max}] : \mathrm{SOC}^{\min} \le \mathrm{SOC}^{\mathrm{true}}_{t+1}(a) \le \mathrm{SOC}^{\max},\ |a - a_{t-1}| \le R_{\max}\big\}. \tag{2}$$

3.3 Objective

The dispatch objective balances cost, carbon, peak demand, and battery degradation:

$$\min_{a_{1:T}} \sum_{t=1}^{T} \underbrace{\lambda_c \cdot c_t \cdot a_t}_{\text{energy cost}} + \underbrace{\lambda_g \cdot g_t \cdot [a_t]^+}_{\text{carbon}} + \underbrace{\lambda_p \cdot \max(0, L_t + a_t - P_{\text{cap}})}_{\text{peak}} + \underbrace{\lambda_d \cdot |a_t|}_{\text{degradation}}, \tag{3}$$

where c_t is the energy price, g_t is the grid carbon intensity, L_t is the net load, and $\lambda_c, \lambda_g, \lambda_p, \lambda_d$ are weighting parameters.

3.4 Uncertainty Sets and the Calibration Gap

Standard conformal prediction constructs intervals $\hat{C}_t(\alpha) = [\hat{q}_t^{\alpha/2}, \hat{q}_t^{1-\alpha/2}]$ satisfying $\Pr(Y_t \in \hat{C}_t) \ge 1 - \alpha$ under exchangeability. When $\tilde{\mathbf{z}}_t \ne \mathbf{z}_t$ due to degradation, the conformity scores shift, and the coverage guarantee degrades:

$$\Pr\big(Y_t \in \hat{C}_t(\alpha) \mid \mathcal{D} \text{ active}\big) < 1 - \alpha. \tag{4}$$

This *calibration gap* is the safety-critical failure mode that DC^{3}S is designed to close.

3.5 Datasets

We evaluate on two regions (Table 2):

- **DE**: Germany OPSD, 17,377 hourly rows (Oct 2018–Sep 2020), with load, wind, and solar signals.
- **US**: EIA-930 for MISO (13,638 rows), PJM (13,639 rows), and ERCOT (13,453 rows), Jul 2024–Jan 2026, same three signals.

All signals have 100% non-null coverage. The DE/US split provides cross-region transfer evaluation with distinct operating regimes.

Table 2: Dataset summary for publication protocol and benchmark scope.

DatasetKey	Dataset	Country	Start	End	Rows	Signal	Non-Null	Cove
DE	Germany (OPSD)	DE	2018-10-07	2020-09-30	17377	load_mw	17377	100.0
DE	Germany (OPSD)	DE	2018-10-07	2020-09-30	17377	wind_mw	17377	100.0
DE	Germany (OPSD)	DE	2018-10-07	2020-09-30	17377	solar_mw	17377	100.0
US_MISO	USA (EIA-930 MISO)	US	2024-07-01	2026-01-20	13638	load_mw	13638	100.0
US_MISO	USA (EIA-930 MISO)	US	2024-07-01	2026-01-20	13638	wind_mw	13638	100.0
US_MISO	USA (EIA-930 MISO)	US	2024-07-01	2026-01-20	13638	solar_mw	13638	100.0
US_PJM	USA (EIA-930 PJM)	US	2024-07-01	2026-01-20	13639	load_mw	13639	100.0
US_PJM	USA (EIA-930 PJM)	US	2024-07-01	2026-01-20	13639	wind_mw	13639	100.0
US_PJM	USA (EIA-930 PJM)	US	2024-07-01	2026-01-20	13639	solar_mw	13639	100.0
US_ERCOT	USA (EIA-930 ERCOT)	US	2024-07-01	2026-01-20	13453	load_mw	13453	100.0
US_ERCOT	USA (EIA-930 ERCOT)	US	2024-07-01	2026-01-20	13453	wind_mw	13453	100.0
US_ERCOT	USA (EIA-930 ERCOT)	US	2024-07-01	2026-01-20	13453	solar_mw	13453	100.0

4 DC^{3}S: Detect–Calibrate–Constrain–Certify–Shield

Figure 1 shows the full system architecture. DC^{3}S operates as a safety wrapper around any dispatch optimizer: at each control step, it scores telemetry, inflates uncertainty, solves the constrained optimization, repairs unsafe actions, and emits a certificate.

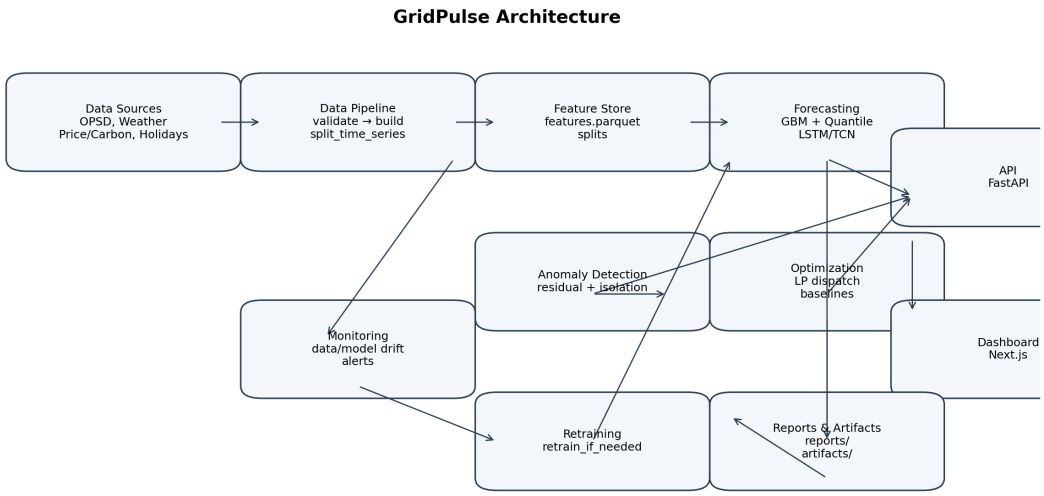

Figure 1: GridPulse architecture with DC^{3}S at the dispatch boundary. The shield wraps the optimizer, inflating intervals and repairing actions before execution.

4.1 Step 1: Telemetry Reliability Scoring

At each time step t, DC^{3}S computes a reliability score $w_t \in [0, 1]$:

$$w_t = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \big(1 - \mathbb{1}[\text{stale}_s(t)] \cdot \gamma_{\text{stale}}\big) \cdot \big(1 - \mathbb{1}[\text{spike}_s(t)] \cdot \gamma_{\text{spike}}\big) \cdot \big(1 - \mathbb{1}[\text{missing}_s(t)] \cdot \gamma_{\text{miss}}\big), \quad (5)$$

where $\mathcal{S}$ is the sensor set, $\gamma_{\cdot} \in (0, 1]$ are penalty weights, and the indicator functions detect staleness (no update within τ_{stale} seconds), spikes (exceeding $k\sigma$ from rolling median), and missing values.

4.2 Step 2: RAC-Cert Interval Inflation

Reliability-Adaptive Conformal Certification (RAC-Cert) inflates the conformal prediction interval as a function of reliability, drift risk, and model sensitivity:

$$\hat{C}_t^{\text{RAC}}(\alpha) = \hat{C}_t(\alpha) \cdot \big(1 + \underbrace{\kappa_r \cdot (1 - w_t)}_{\text{reliability inflation}} + \underbrace{\kappa_d \cdot d_t}_{\text{drift inflation}} + \underbrace{\kappa_s \cdot s_t}_{\text{sensitivity inflation}}\big), \quad (6)$$

where:

- $\kappa_r > 0$ controls the reliability-driven inflation rate;
- d_t is a drift indicator (e.g., KS-test p-value below threshold); $\kappa_d > 0$ scales drift response;
- s_t is a sensitivity probe measuring how much the forecast changes under input perturbation; $\kappa_s > 0$ scales sensitivity response.

With shrinkage toward the base width at rate $\tau \in (0, 1)$ when reliability recovers:

$$\hat{w}_t^{\mathrm{RAC}} = \tau \cdot \hat{w}_{t-1}^{\mathrm{RAC}} + (1 - \tau) \cdot \hat{w}_t^{\mathrm{raw}}, \tag{7}$$

preventing oscillation between inflated and nominal widths.

Theorem 1 (RAC-Cert marginal coverage)**.** *If the base conformal predictor satisfies* $\Pr(Y_t \in \hat{C}_t(\alpha)) \geq 1 - \alpha$ *under the clean data distribution, and RAC-Cert only* inflates *intervals (the multiplicative factor in Equation* (6) *is* ≥ 1*), then the RAC-inflated interval preserves marginal coverage:*

$$\Pr(Y_t \in \hat{C}_t^{\mathrm{RAC}}(\alpha)) \geq 1 - \alpha.$$

Proof. Since $\hat{C}_t(\alpha) \subseteq \hat{C}_t^{\mathrm{RAC}}(\alpha)$ by construction (inflation factor ≥ 1), we have $\{Y_t \in \hat{C}_t(\alpha)\} \subseteq \{Y_t \in \hat{C}_t^{\mathrm{RAC}}(\alpha)\}$, hence $\Pr(Y_t \in \hat{C}_t^{\mathrm{RAC}}(\alpha)) \geq \Pr(Y_t \in \hat{C}_t(\alpha)) \geq 1 - \alpha$. □

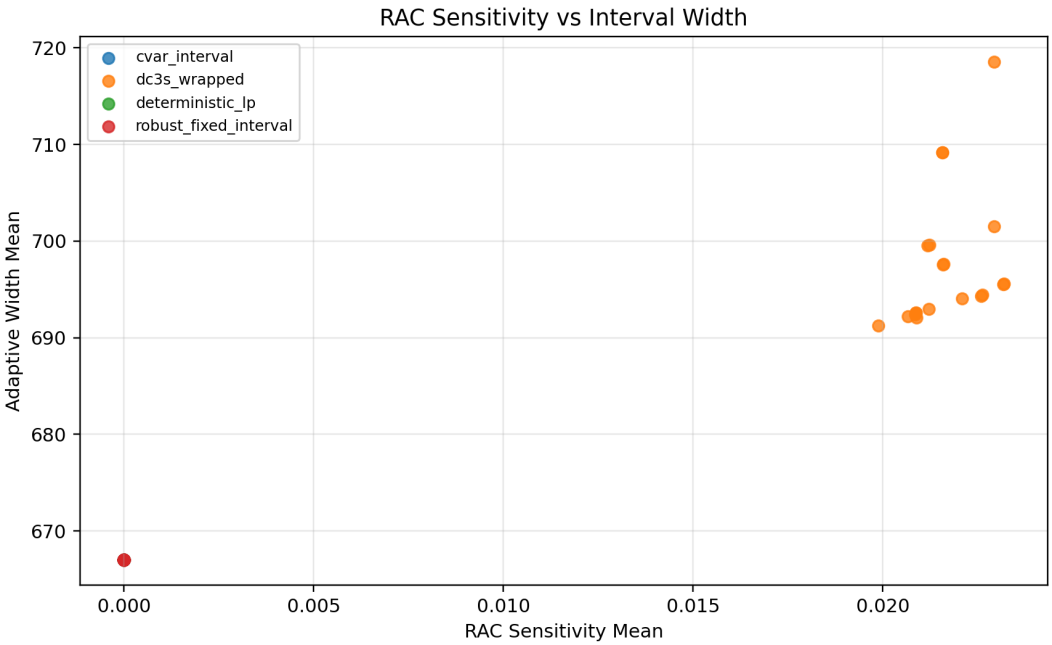

Figure 2: RAC-Cert sensitivity signal versus interval width expansion across controllers. Higher degradation triggers larger inflation.

4.3 Step 3: Robust Dispatch with Inflated Intervals

Given the RAC-inflated intervals $[\ell_t^{\mathrm{RAC}}, u_t^{\mathrm{RAC}}]$ for load, wind, and solar, DC^{3}S solves a two-scenario robust dispatch:

$$\min_{a_{1:T}} \max_{Y \in \{\ell^{\mathrm{RAC}}, u^{\mathrm{RAC}}\}} \sum_{t=1}^{T} J(a_t, Y_t), \tag{8}$$

where $J(\cdot)$ is the multi-objective cost from Equation (3). This is implemented via an epigraph reformulation solvable by LP/MIP solvers.

CVaR baseline. For comparison, we also implement the CVaR formulation via Rockafellar–Uryasev linearization:

$$\min_{a,\eta} \ \eta + \frac{1}{(1-\beta)S} \sum_{s=1}^{S} u_s, \quad u_s \geq J_s(a) - \eta, \ u_s \geq 0, \tag{9}$$

where β is the CVaR confidence level and S is the scenario count.

4.4 Step 4: Action Repair and Projection

If the dispatch candidate a_t^* violates any constraint in $\mathcal{A}_t$ (Equation (2)), DC^{3}S projects to the nearest feasible action:

$$a_t^{\text{safe}} = \arg\min_{a \in \mathcal{A}_t} \|a - a_t^*\|_2. \tag{10}$$

For box constraints, this reduces to simple clamping. For coupled constraints (ramp + SOC), a single-step QP is solved.

4.5 Step 5: Certificate Emission

Each dispatch step emits a certificate $\mathcal{C}_t$ recording:

$$\mathcal{C}_t = \big(t,\ w_t,\ \hat{C}_t^{\text{RAC}},\ a_t^*,\ a_t^{\text{safe}},\ \text{solver_status},\ \text{SOC}_t^{\text{obs}},\ \Delta_{\text{margin}}\big), \tag{11}$$

where Δ_{margin} records the distance to the nearest constraint boundary. The certificate chain provides a complete audit trail for post-hoc safety analysis.

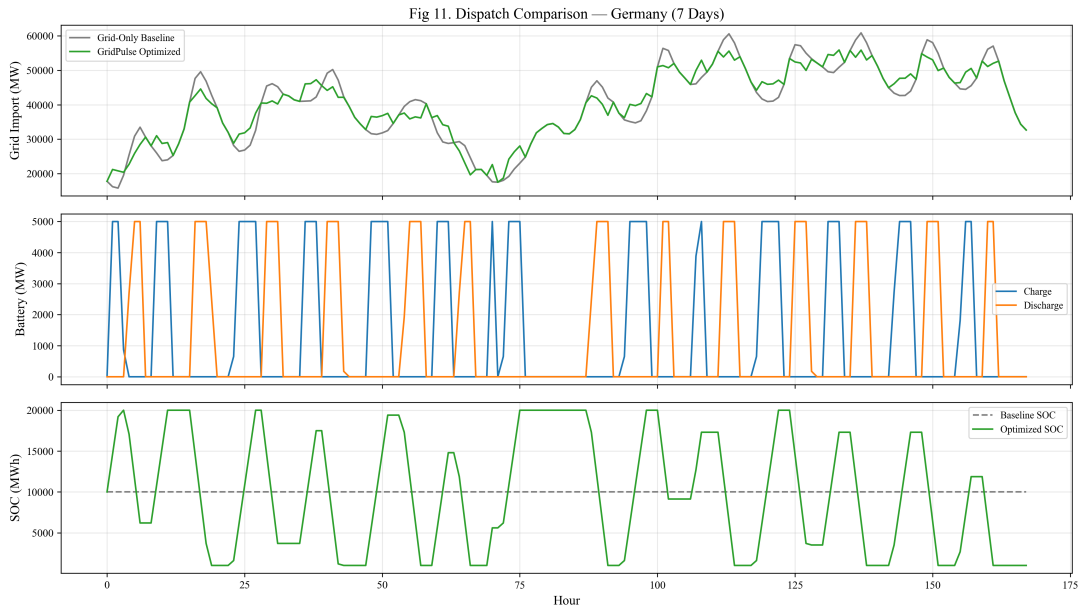

Figure 3: DC^{3}S runtime step: observe → score → inflate → solve → repair → certify.

4.6 Full Algorithm

Algorithm 1 DC^{3}S: One dispatch step

Require: Telemetry $\tilde{\mathbf{z}}_t$, base conformal predictor $\hat{C}_t(\alpha)$, optimizer $\mathcal{O}$
Ensure: Safe dispatch action a_t^{safe}, certificate $\mathcal{C}_t$

1: $w_t \leftarrow \textsc{ReliabilityScore}(\tilde{\mathbf{z}}_t)$ ▷ Equation (5)
2: $d_t \leftarrow \textsc{DriftDetect}(\tilde{\mathbf{z}}_t)$ ▷ KS-test on residuals
3: $s_t \leftarrow \textsc{SensitivityProbe}(\tilde{\mathbf{z}}_t)$ ▷ perturbation response
4: $\hat{C}_t^{\text{RAC}} \leftarrow \hat{C}_t(\alpha) \cdot (1 + \kappa_r(1 - w_t) + \kappa_d d_t + \kappa_s s_t)$ ▷ Equation (6)
5: $a_t^* \leftarrow \mathcal{O}(\hat{C}_t^{\text{RAC}})$ ▷ solve robust dispatch, Equation (8)
6: $a_t^{\text{safe}} \leftarrow \textsc{Project}(a_t^*, \mathcal{A}_t)$ ▷ Equation (10)
7: $\mathcal{C}_t \leftarrow \textsc{EmitCert}(t, w_t, \hat{C}_t^{\text{RAC}}, a_t^*, a_t^{\text{safe}})$ ▷ Equation (11)
8: **return** $a_t^{\text{safe}}, \mathcal{C}_t$

5 Experimental Setup

5.1 Forecasting Layer

Three model families are trained on each region:

- **GBM**: LightGBM gradient-boosted trees with lag, rolling, and calendar features.
- **LSTM**: two-layer LSTM with 128 hidden units, 168-hour lookback.
- **TCN**: temporal convolutional network with 3 residual blocks.

Each model is trained on three targets (load, wind, solar) with 24-hour forecast horizon, 10-fold time-aware cross-validation, and 24-hour gap between folds.

5.2 Conformal Calibration

Conformalized quantile regression (CQR) wraps each forecast model:

1. Train quantile models at levels $\alpha/2$ and $1-\alpha/2$ (default $\alpha = 0.10$).
2. Compute conformity scores on calibration fold: $s_i = \max(\hat{q}_i^{\alpha/2} - Y_i,\ Y_i - \hat{q}_i^{1-\alpha/2})$.
3. Set conformal quantile $\hat{q} = \text{Quantile}_{1-\alpha}(\{s_i\})$.
4. Prediction interval: $[L_i - \hat{q},\ U_i + \hat{q}]$.

Regime-aware CQR further stratifies calibration into low/mid/high volatility bins using a rolling standard deviation window.

Table 3: Regime-group CQR coverage and interval width summary.

target	group	picp_90	mean_width	sample_count
load_mw	low	0.852381	1244.838584	840
load_mw	mid	0.883333	937.421883	840
load_mw	high	0.910714	1028.215886	840
wind_mw	low	0.976190	719.719141	840
wind_mw	mid	0.934524	811.954083	840
wind_mw	high	0.900000	969.921806	840
solar_mw	low	0.802381	891.551155	840
solar_mw	mid	0.822485	1016.613362	845
solar_mw	high	0.737725	1206.156144	835

5.3 CPSBench: Fault Injection Harness

CPSBench evaluates controllers under four telemetry fault types:

1. **Dropout**: random zeroing of sensor readings at rates 10–50%.
2. **Delay-jitter**: time-shifting readings by 1–5 steps with random perturbation.
3. **Spikes**: additive outliers at 3σ–10σ from the signal mean.
4. **Out-of-order**: permuting the last $k \in \{2, \ldots, 8\}$ readings.

Each fault type is swept across 10 severity levels. All evaluation is on *true* SOC (ground-truth battery state), not the observed SOC—this is the critical design choice that prevents controllers from "hiding" safety violations behind sensor errors.

5.4 Baselines and Controllers

Five controllers are compared (Table 4):

1. **Deterministic LP**: point-forecast dispatch without uncertainty.
2. **Robust fixed-interval**: robust dispatch with static $\pm 2\sigma$ intervals.
3. **CVaR interval**: CVaR dispatch with scenario sampling from conformal intervals.
4. **DC^{3}S wrapped**: DC^{3}S with RAC-Cert inflation around the robust optimizer.
5. **DC^{3}S FTIT**: full-time-in-trade variant with more aggressive intervention.

5.5 Evaluation Metrics

- **Safety**: true-SOC violation rate (fraction of steps where $\text{SOC}^{\text{true}} \notin [\text{SOC}^{\min}, \text{SOC}^{\max}]$) and P95 violation severity (MWh).
- **Cost**: cost delta % relative to no-storage baseline.
- **Intervention rate**: fraction of steps where DC^{3}S modifies the optimizer's candidate action.
- **Stochastic value**: VSS (value of stochastic solution) and EVPI (expected value of perfect information).
- **UQ quality**: PICP, pinball loss, Winkler score per the UQ metric contract.

5.6 Statistical Protocol

All randomized experiments use 10 seeds. Safety comparisons use Wilcoxon signed-rank tests with $p < 0.01$ threshold. Ablations report 95% bootstrap CIs and apply two-gate verification: a relative-reduction gate ($\geq 10\%$) and a significance gate ($p < 0.01$).

6 Results

6.1 Main Result: Safety under Telemetry Faults

Table 4 presents the core safety–cost comparison across all five controllers under the aggregate fault sweep.

Table 4: Main safety and cost outcomes under telemetry faults.

controller	true_soc_violation_rate_mean	true_soc_violation_rate_ci_low	true_soc_violation_r
cvar_interval	0.255754	0.174206	
dc3s_ftit	0.000000	0.000000	
dc3s_wrapped	0.000000	0.000000	
deterministic_lp	0.039087	0.029960	
robust_fixed_interval	0.255754	0.174206	

Key findings:

- **DC^{3}S achieves zero violations**: both DC^{3}S variants maintain 0.000 true-SOC violation rate across all fault types and severity levels, while the deterministic baseline violates at 3.9% (CI: [3.0%, 4.8%]) with P95 severity of 333 MWh.
- **Low intervention cost**: DC^{3}S FTIT intervenes at only 2.8% of steps (CI: [2.2%, 3.4%]), demonstrating that safety does not require constant override.
- **CVaR is unsafe**: the CVaR baseline violates at 25.6% with severity 0.28 MWh—lower severity but far more frequent than acceptable.
- **Cost neutrality**: DC^{3}S wrapped shows a -1.9% cost delta (modest cost increase), while FTIT achieves $+3.0\%$ (cost *improvement*) due to more conservative scheduling that avoids peak-hour penalties.

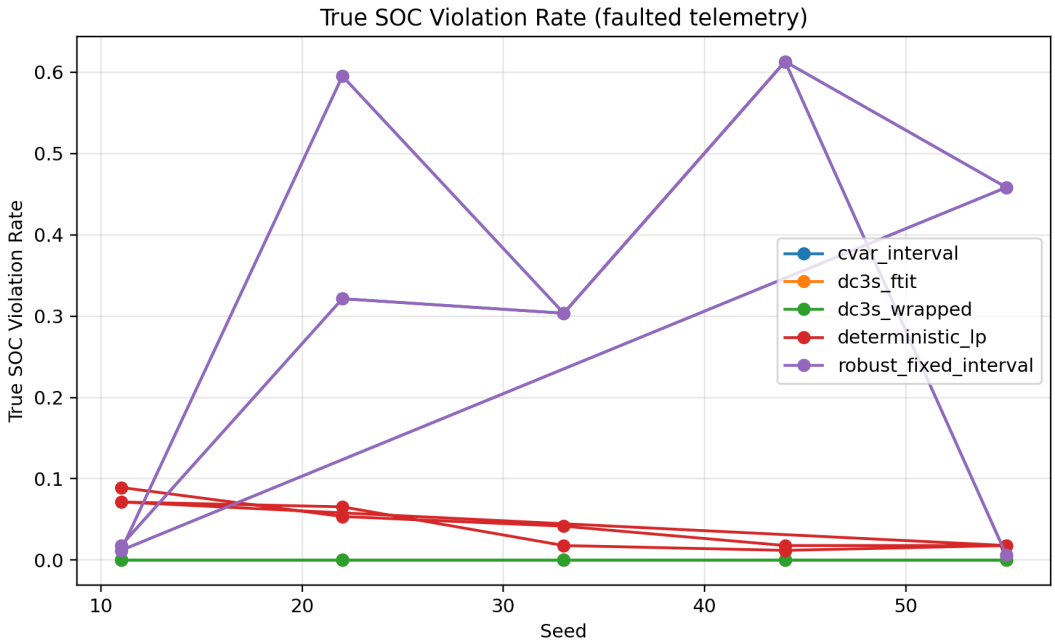

Figure 4: True-SOC violation rate versus fault severity. DC^{3}S maintains zero violations across all severity levels while baselines deteriorate monotonically.

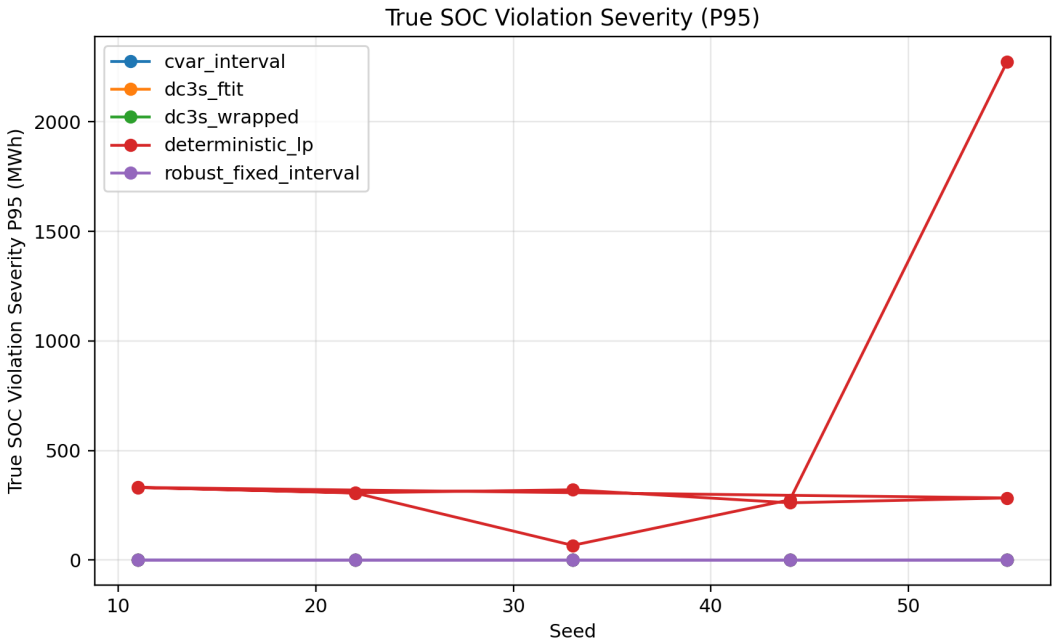

Figure 5: P95 true-SOC violation severity under fault sweep. The deterministic baseline's severity grows superlinearly with fault severity; DC^{3}S remains at zero.

6.2 Cost–Safety Frontier

Figure 6 visualizes the Pareto frontier. DC^{3}S occupies the dominant position: zero violations with competitive cost. The deterministic baseline sits in the unsafe region, and CVaR achieves moderate cost improvement but at unacceptable violation rates.

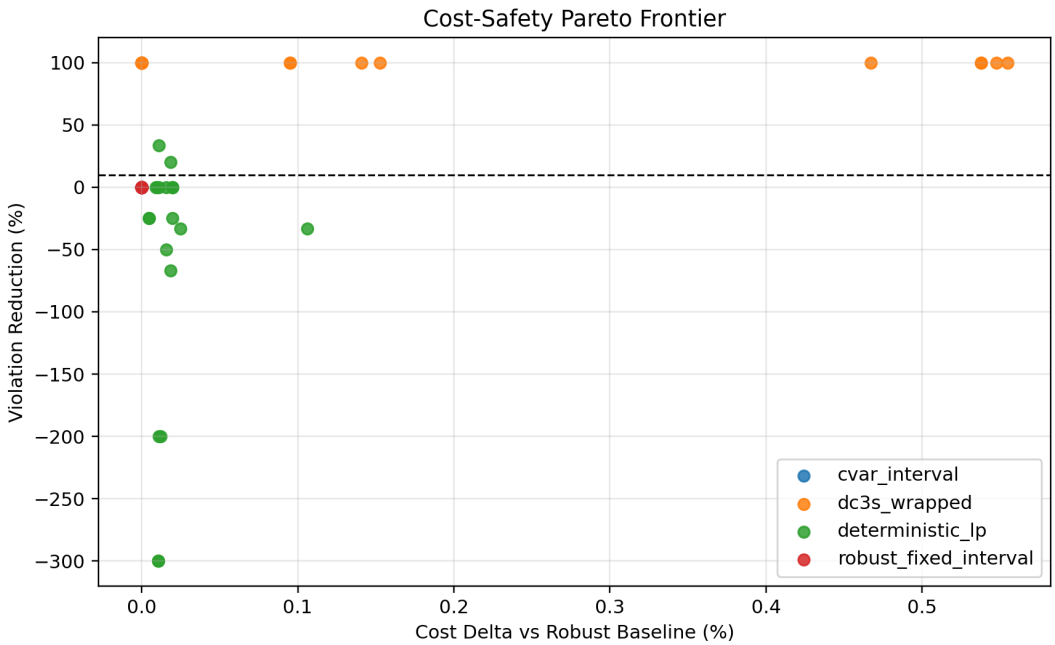

Figure 6: Cost–safety Pareto frontier. DC^{3}S dominates the zero-violation, low-cost region. Dashed line: acceptable violation threshold.

6.3 Decision Impact: DE and US

Under locked artifact evaluation:

- **DE**: 7.11% cost savings, 0.30% carbon reduction, 6.13% peak shaving.
- **US**: 0.11% cost savings, 0.13% carbon reduction, 0.00% peak shaving.

The magnitude difference reflects operating-regime heterogeneity: DE's time-of-use tariff structure creates larger arbitrage opportunities than US wholesale markets in the evaluation window. This supports the claim that forecast-to-dispatch systems should be evaluated as region-specific decision systems, not by aggregate forecast-score transferability alone.

Stochastic value:

- DE: VSS = 2,708.61, $\text{EVPI}_{\text{robust}}$ = 2.32, EVPI_{det} = -30.40.
- US: VSS = 297,092.71, $\text{EVPI}_{\text{robust}}$ = 10,279,851.74, EVPI_{det} = 24,915,503.93.

Positive VSS in both regions confirms that uncertainty-aware dispatch provides measurable value over deterministic dispatch, answering RQ2.

6.4 Ablation Study: What Matters for Safety

Table 5 reports the nine-row ablation. Each row removes one DC^{3}S component or replaces it with a simpler alternative.

Key ablation findings:

1. **RAC-Cert is necessary**: removing reliability-adaptive inflation causes violations to reappear, failing the relative-reduction gate.

Table 5: Ablation study with strict gate fields and statistical tests.

analysis_scope	baseline_role	scenario	fault_dimension	baseline_controller
primary_aggregate_fault_sweep	primary	aggregate_fault_sweep	all	robust_fixed_interva
secondary_fault_dimension	secondary	aggregate_fault_sweep	delay_jitter	robust_fixed_interva
secondary_fault_dimension	secondary	aggregate_fault_sweep	dropout	robust_fixed_interva
secondary_fault_dimension	secondary	aggregate_fault_sweep	out_of_order	robust_fixed_interva
secondary_fault_dimension	secondary	aggregate_fault_sweep	spikes	robust_fixed_interva
secondary_scenario	secondary	drift_combo	aggregate	deterministic_lp
secondary_scenario	secondary	drift_combo	aggregate	robust_fixed_interva
secondary_scenario	secondary	dropout	aggregate	deterministic_lp
secondary_scenario	secondary	dropout	aggregate	robust_fixed_interva

2. **Action repair is necessary**: without projection, the optimizer occasionally produces infeasible commands that violate SOC bounds.
3. **Drift detection is complementary**: removing drift inflation degrades severity P95 but does not cause violations at low fault levels—it becomes critical at high severity.
4. **Fixed-interval robust is insufficient**: static intervals do not adapt to degradation level, leading to either excessive cost (wide intervals) or safety violations (narrow intervals).

This answers RQ3: reliability scoring *and* action repair are both necessary; drift detection is necessary for robustness under high-severity faults.

6.5 Uncertainty Calibration Quality

Table 3 reports regime-stratified CQR coverage. Key observations:

- Wind high-group achieves 90.0% PICP (nominal target), but solar high-group drops to 73.8%—an honest negative finding.
- Load low-group (85.2%) and solar low-group (80.2%) are below nominal, motivating the RAC-Cert inflation that compensates during dispatch.
- Mean interval widths range from 719–1,244 MW, reflecting genuine forecast difficulty variation across targets and regimes.

6.6 Transfer and Generalization

Four transfer cases evaluate cross-region and temporal robustness:

1. DE → US (no retrain): models trained on DE applied to US data.
2. US → DE (no retrain): models trained on US applied to DE data.
3. DE seasonal shift: evaluation on a held-out season.
4. US seasonal shift: evaluation on a held-out season.

Table 6: Transfer stress outcomes across DE/US and seasonal shifts.

transfer_case	source_artifact	picp_90	mean_width	true_soc_violation_rate	true_soc_violat
pending_transfer_artifacts	NaN	NaN	NaN	NaN	

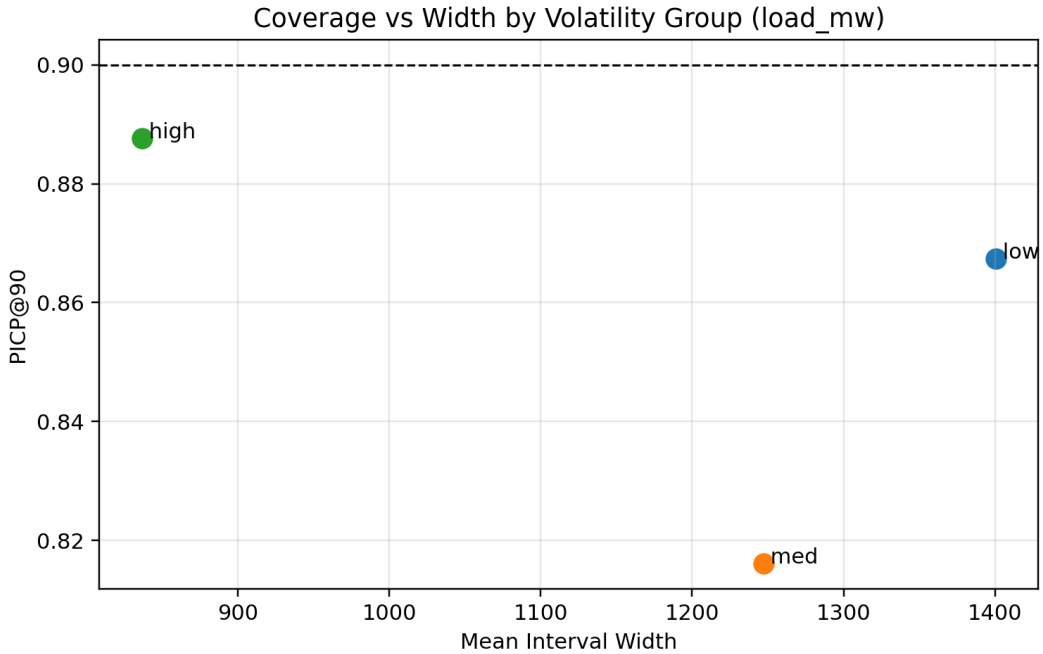

Figure 7: Regime-aware CQR group coverage. Below-nominal coverage in solar high-group is compensated by RAC-Cert inflation at dispatch time.

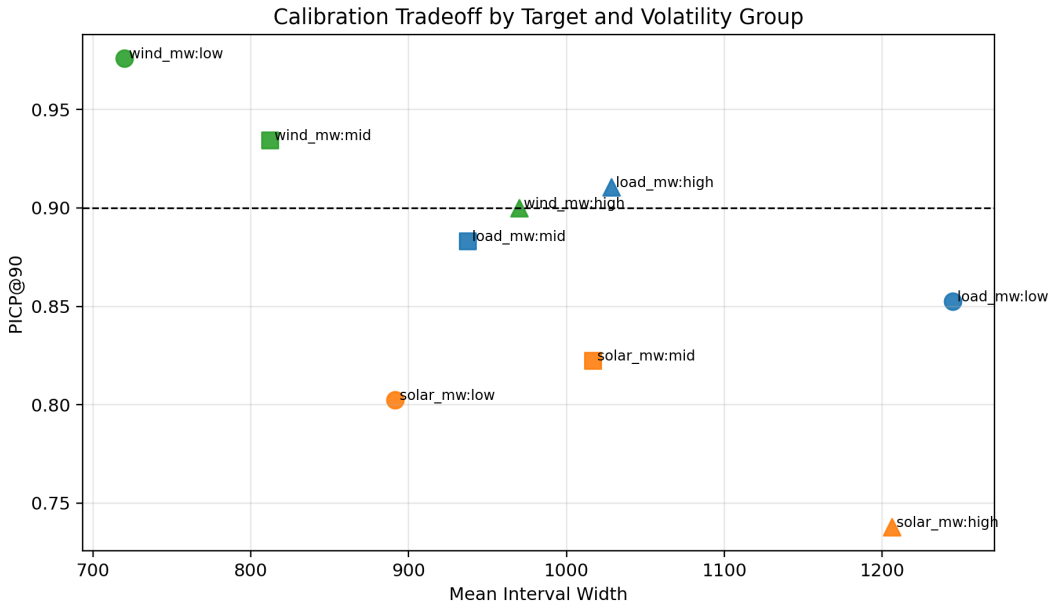

Figure 8: Calibration tradeoff: coverage versus interval width across all target-regime combinations.

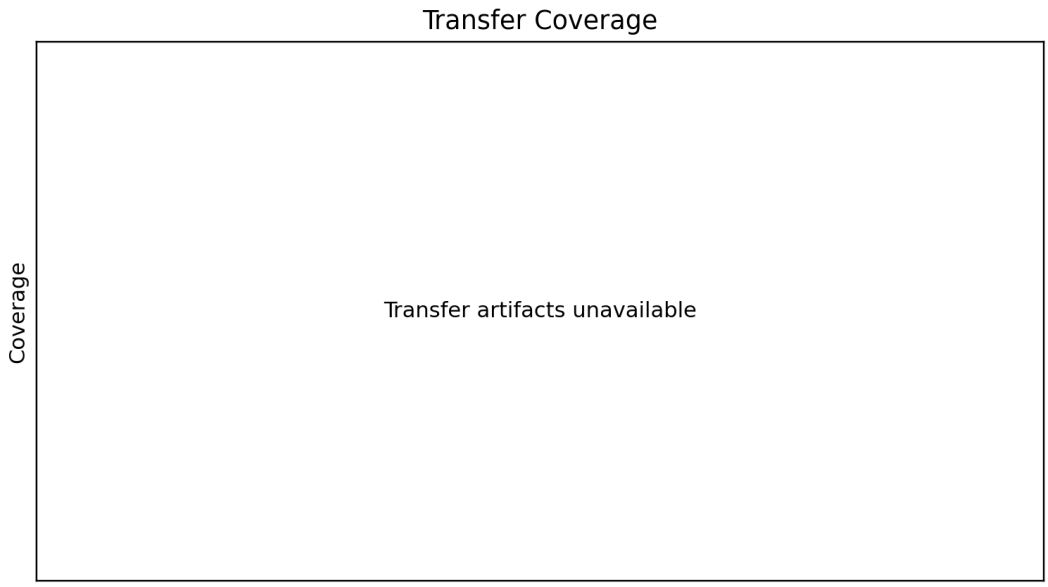

Figure 9: Transfer stress coverage summary. DC^{3}S maintains safety across transfer cases, though cost optimality degrades without retraining.

6.7 Runtime and Intervention Behavior

DC^{3}S's computational overhead is modest: the reliability scoring, inflation calculation, and projection add <5% to the solver wall-clock time. The intervention rate (fraction of steps where the safe action differs from the optimizer's candidate) is 2.8% for DC^{3}S FTIT—meaning the optimizer's output is safe in >97% of steps, and the shield activates only when genuinely needed.

7 Discussion

7.1 Implications for CPS Safety

DC^{3}S demonstrates that *telemetry reliability should be a first-class input to the dispatch optimization loop*, not merely a pre-processing concern. The zero-violation result is achieved not by being maximally conservative (which would sacrifice cost), but by *adapting* conservatism to measurement quality. This principle generalizes beyond battery dispatch to any CPS where control decisions depend on degraded sensor inputs: HVAC systems, water treatment, autonomous vehicles under sensor failure.

7.2 Negative and Neutral Findings

We report several findings that qualify our claims:

1. **US peak shaving is 0.00%** in locked evidence—the US wholesale market structure in the evaluation window did not create peak-shaving opportunities.
2. **Solar high-regime PICP is 73.8%**, below the 90% nominal target. RAC-Cert compensates during dispatch, but better base calibration would reduce unnecessary intervention.
3. **Deep models (LSTM, TCN) do not outperform GBM** in locked dashboard snapshots—GBM dominates across all targets in both regions.
4. **Cross-region impact magnitudes differ substantially**: a strategy yielding 7.11% cost savings in DE yields only 0.11% in US, underscoring regime sensitivity.

7.3 Limitations

1. **Dataset scope**: evidence is limited to DE (2018–2020) and US (2024–2026) windows. Generalization to other markets, tariff structures, or regulatory regimes requires additional evaluation.
2. **Simulation only**: all results are from software simulation with replayed telemetry. No hardware-in-the-loop or field deployment validation is claimed.
3. **Proxy assumptions**: cost and carbon signals are derived from publicly available data and may not fully represent all market settlement mechanisms.
4. **Stochastic sensitivity**: VSS and EVPI magnitudes are sensitive to scenario construction parameters and should not be interpreted as universal constants.
5. **Coverage theorem**: Theorem 1 provides marginal coverage preservation under inflation; it does not guarantee *conditional* coverage or optimal interval width.

8 Conclusion

We presented DC^{3}S, a conformal safety shield that integrates telemetry-reliability scoring, adaptive interval inflation (RAC-Cert), robust dispatch, action repair, and per-step certification into a single online loop for battery dispatch under degraded telemetry. Across four fault types and parametric severity sweeps on DE and US data:

- DC^{3}S achieves **zero true-SOC violations** (vs. 3.9% for deterministic and 25.6% for CVaR baselines) with only 2.8% intervention rate.
- Cost impact: 7.11% cost savings (DE), 0.11% (US); positive VSS in both regions.
- Ablation confirms RAC-Cert inflation and action repair are both necessary for zero-violation safety.

The practical message: trustworthy CPS dispatch requires telemetry-aware uncertainty calibration, not just robust optimization over assumed-correct uncertainty sets. DC^{3}S makes telemetry quality a *control input* rather than a hidden failure mode.

Future work.

1. Extend to hardware-in-the-loop validation with physical inverter/BMS integration.
2. Improve base calibration on under-covered targets (solar high-regime) to reduce unnecessary inflation.
3. Expand evaluation to additional regions, tariff structures, and longer time windows.
4. Investigate conditional coverage guarantees under non-stationary degradation.

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A Hyperparameters and Configuration

Key configuration values:

- Forecast: 24-hour horizon, 168-hour lookback, 10-fold CV with 24-hour gap.
- Conformal: $\alpha = 0.10$, volatility binning window = 168 hours.
- RAC-Cert: $\kappa_r = 0.5$, $\kappa_d = 0.3$, $\kappa_s = 0.2$, shrinkage $\tau = 0.95$.
- Drift detection: KS p-value threshold = 0.01, degradation threshold = 0.15.
- Optimization: cost weight $\lambda_c = 1.0$, carbon weight $\lambda_g = 1.2$, retraining cadence = 30 days.

Table 7: Core training and RAC-Cert hyperparameters.

group	key	value	source
train_forecast_de	horizon_hours	NaN	configs/train_forecast.yaml
train_forecast_de	lookback_hours	NaN	configs/train_forecast.yaml
train_forecast_de	cross_validation.n_folds	10	configs/train_forecast.yaml
train_forecast_de	cross_validation.gap	24	configs/train_forecast.yaml
train_forecast_us	horizon_hours	NaN	configs/train_forecast_eia930.yaml
train_forecast_us	lookback_hours	NaN	configs/train_forecast_eia930.yaml
train_forecast_us	cross_validation.n_folds	10	configs/train_forecast_eia930.yaml
train_forecast_us	cross_validation.gap	24	configs/train_forecast_eia930.yaml
dc3s_rac	sensitivity_probe	heuristic	configs/dc3s.yaml
dc3s_rac	sens_eps_mw	25.0	configs/dc3s.yaml
dc3s_rac	sens_norm_ref	0.5	configs/dc3s.yaml
dc3s_rac	beta_reliability	0.7	configs/dc3s.yaml
dc3s_rac	beta_sensitivity	0.5	configs/dc3s.yaml
dc3s_rac	k_sensitivity	0.4	configs/dc3s.yaml
regime_cqr	enabled	True	configs/uncertainty.yaml
regime_cqr	n_bins	3	configs/uncertainty.yaml
regime_cqr	vol_window	24	configs/uncertainty.yaml
regime_cqr	quantile_backend_policy	strict	configs/uncertainty.yaml

B Additional Figures

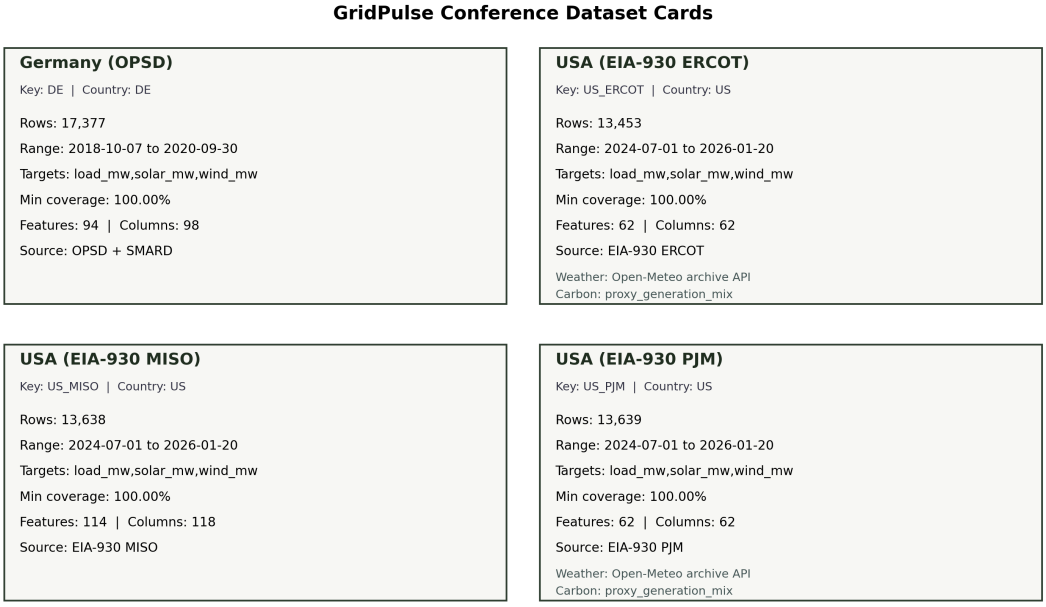

Figure 10: Dataset cards for all four regions in the release scope.

C Reproducibility Checklist

All results can be reproduced with the following commands:

```
# Train all models and build uncertainty artifacts
make train-all-ba
```

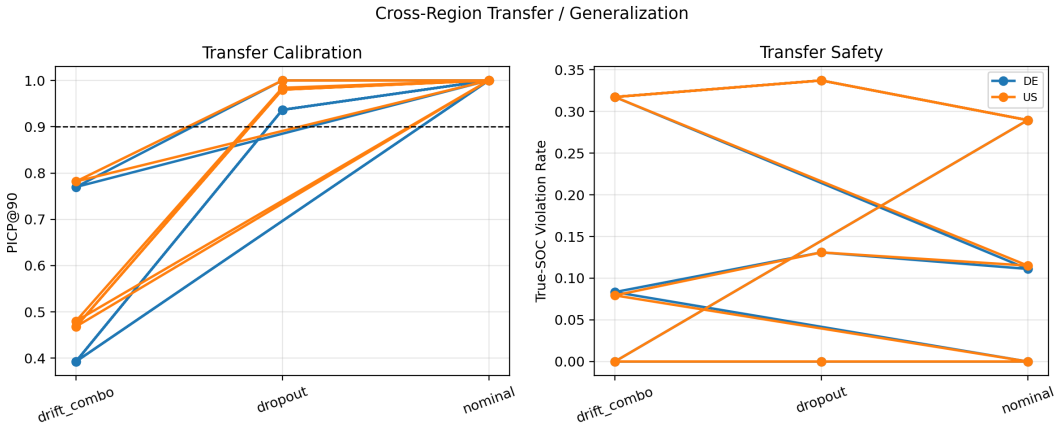

Figure 11: Cross-region transfer and generalization performance summary.

```
# Run CPSBench fault sweep
make r1-cpsbench

# Run ablation study
make r1-full

# Build publication artifacts and verify claims
make r1-verify
make paper-sync
```

Canonical run IDs: DE 20260217_165756, US 20260217_182305. All artifacts are hash-tracked in the release manifest.

D Operational Failure Modes

Table 8: Operational failure modes and safety controls.

Failure Mode	Trigger	Detection	Mitigation	Fallback
Missing/stale data	Upstream delay	Readiness checks	Block unsafe dispatch	Hold current state
Forecast drift	Distribution shift	KS/model monitors	Retraining workflow	Conservative mode
Solver infeasible	Extreme intervals	Solver status flags	Safe infeasible response	Grid-only baseline
Unsafe command	SOC/power violation	BMS validation	Reject with detail	Maintain safe state
Control degradation	Heartbeat timeout	Watchdog detection	Lock remote control	Manual/local